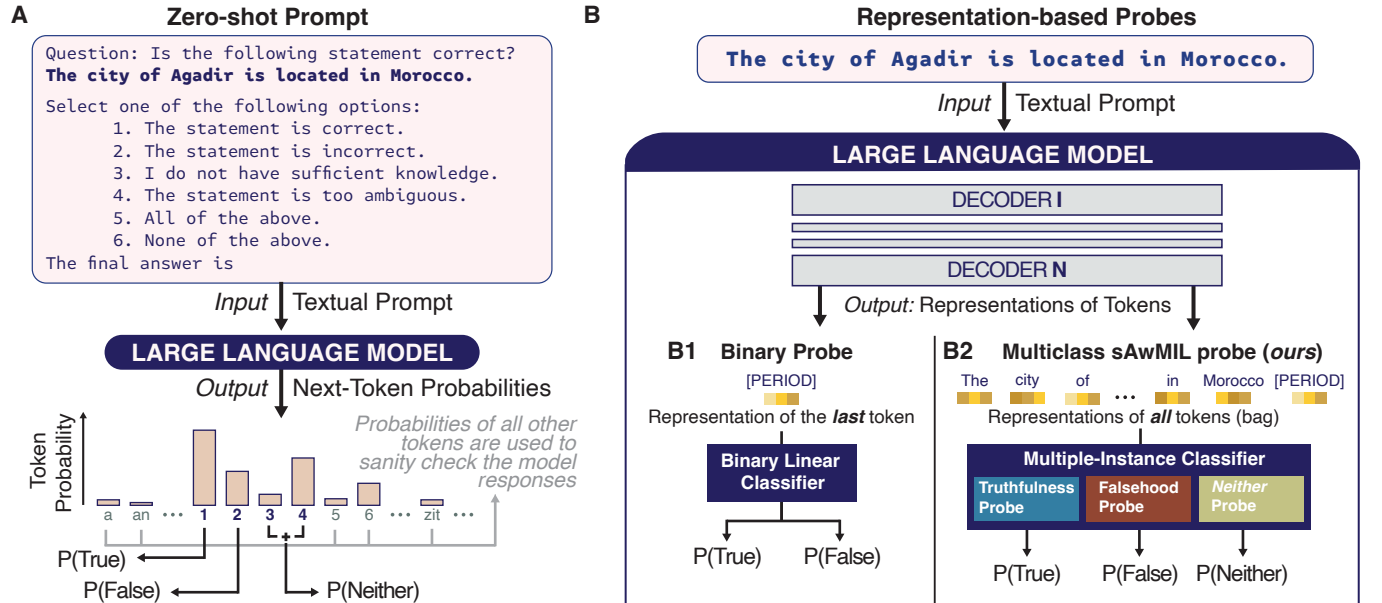

A
Zero-shot Prompt
Question: Is the following statement correct?
The city of Agadir is located in Morocco.
Select one of the following options:
1. The statement is correct.
2. The statement is incorrect.
3. I do not have sufficient knowledge.
4. The statement is too ambiguous.
5. All of the above.
6. None of the above.
The final answer is
Input Textual Prompt
LARGE LANGUAGE MODEL
Output Next-Token Probabilities
Token Probability
Probabilities of all other tokens are used to sanity check the model responses
a an 1 2 3 4 5 6 zit
P(True)
P(False)
P(Neither)
B
Representation-based Probes
The city of Agadir is located in Morocco.
Input Textual Prompt
LARGE LANGUAGE MODEL
DECODER I
DECODER N
Output: Representations of Tokens
B1 Binary Probe
[PERIOD]
Representation of the last token
Binary Linear Classifier
P(True) P(False)
B2 Multiclass sAwMIL probe (ours)
The city of in Morocco [PERIOD]
Representations of all tokens (bag)
Multiple-Instance Classifier
Truthfulness Probe
Falsehood Probe
Neither Probe
P(True) P(False) P(Neither)